THE MIDDLE FACTOR OF A COMPOSITE IS OF FINITE PRESENTATION WHEN THE COMPOSITE IS AND THE SECOND FACTOR IS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The definition of a morphism of algebraic stacks being locally of finite presentation through smooth covers by algebraic spaces. The author's section on morphisms of finite presentation, in which the counted statement occurs. Reduces the assertion about stacks to an assertion about morphisms of algebraic spaces.
- The corresponding permanence statement for morphisms of algebraic spaces. Invoked at the end of the counted proof (source lines 5697-5700) as a cross-chapter pointer. Provides the statement to which the proof reduces.

Source reference. Original reference spaces-morphisms-lemma-finite-presentation-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5699–5699. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-finite-presentation-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Y} \to \mathcal{Z}$ be morphisms of algebraic stacks If $g \circ f$ is locally of finite presentation and g is locally of finite type, then f is locally of finite presentation.*

Proof. Choose an algebraic space W and a surjective smooth morphism $W \to \mathcal{Z}$. Choose an algebraic space V and a surjective smooth morphism $V \to \mathcal{Y} \times_{\mathcal{Z}} W$. Choose an algebraic space U and a surjective smooth morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$.

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The lemma follows upon applying Morphisms of Spaces, Lemma spaces-morphisms-lemma-finite-presentation-permanence to the morphisms $U \to V \to W$. $\square$